[CSL252]-Design and Analysis of Algorithms **2025-26-W**

TUTORIAL XII

Date of the tutorial: **Apr 09, 2026**.

Due Date: **11:59 pm, Apr 13, 2026**.

Instructions. Same as Tutorial I

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1. This is a reading exercise. Study at least twelve examples of polynomial-time reductions (including those discussed in class). List the reductions you have studied, using standard notation. For example: (a) Vertex Cover $\leq_p$ Independent Set. You just need list twelve reductions you have studied.

2. Consider the following heuristic for constructing a vertex cover of a connected graph G: return the set of non-leaf nodes in any depth-first spanning tree of G.

   (a) Prove that this heuristic returns a vertex cover of G.

   (b) Prove that this heuristic returns a 2-approximation to the minimum vertex cover of G.

   (c) Describe an infinite family of graphs for which this heuristic returns a vertex cover of size $2 \cdot OPT$.

3. Given a undirected graph $G = (V, E)$, we say that a triple (u, v, w) forms a triangle if $uv, vw, uw \in E(G)$. Design an algorithm that, given a graph G, returns a set of vertices S (as small as possible) such that $G \setminus S$ does not contain any triangle.

   (a) Describe an integer program for the problem and corresponding LP relaxation.

   (b) (Rounding) Describe a rounding procedure based on rounding procedure for vertex cover.

   (c) (Correctness) Let S' be the output of the rounding procedure. Prove that $G \setminus S'$ does not contain any triangle.

   (d) (Approximation) Analyze the apporximation guarantee.

   (e) (Runtime) What is the running time of the algorithm?

4. Recall that in the basic Load Balancing Problem, we are interested in assigning jobs to machines so as to minimize the makespan, i.e., the maximum load on any machine.

   In many applications, machines may have different processing speeds. Consider the following model.

   Suppose you are given a system consisting of m slow machines and k fast machines. Each fast machine can process jobs at twice the speed of a slow machine. There are n jobs, and each job i requires time t_i when processed on a slow machine and time $\frac{t_i}{2}$ when processed on a fast machine.

   The goal is to assign each job to a machine so as to minimize the makespan, defined as the maximum total processing time assigned to any machine.

   Question: Give a polynomial-time algorithm that produces an assignment of jobs to machines with makespan at most 3 times the optimal makespan.